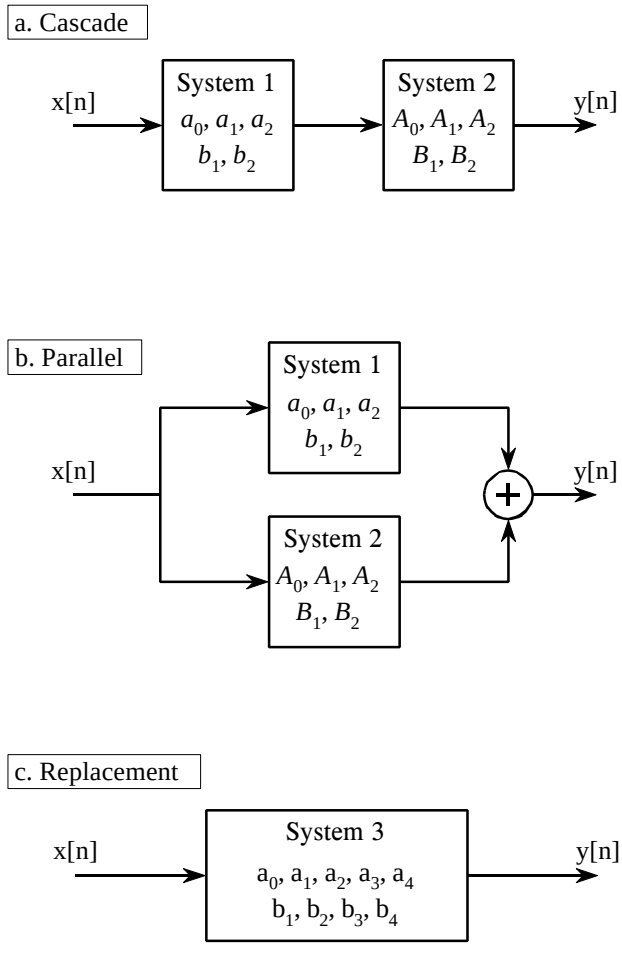

FIGURE 33-4
Combining cascade and parallel stages. The z-domain allows recursive stages in a cascade, (a), or in parallel, (b), to be combined into a single system, (c).

Since this is in the form of Eq. 33-3, we can directly extract the recursion coefficients that implement the cascaded system:

$$
\begin{aligned}
a_0 &= a_0 A_0 & & \\
a_1 &= a_0 A_1 + a_1 A_0 & b_1 &= b_1 + B_1 \\
a_2 &= a_0 A_2 + a_1 A_1 + a_2 A_0 & b_2 &= b_2 + B_2 - b_1 B_1 \\
a_3 &= a_1 A_2 + a_2 A_1 & b_3 &= -b_1 B_2 - b_2 B_1 \\
a_4 &= a_2 A_2 & b_4 &= -b_2 B_2
\end{aligned}
$$

The obvious problem with this technique is the large amount of algebra needed to multiply and rearrange the polynomial terms. Fortunately, the entire algorithm can be expressed in a short computer program, shown in Table 33-1. Although the cascade and parallel combinations require different mathematics, they use nearly the same program. In particular, only one line of code is different between the two algorithms, allowing both to be combined into a single program.